Programming a RTL-SDR with MATLAB

Classes and objects in MATLAB

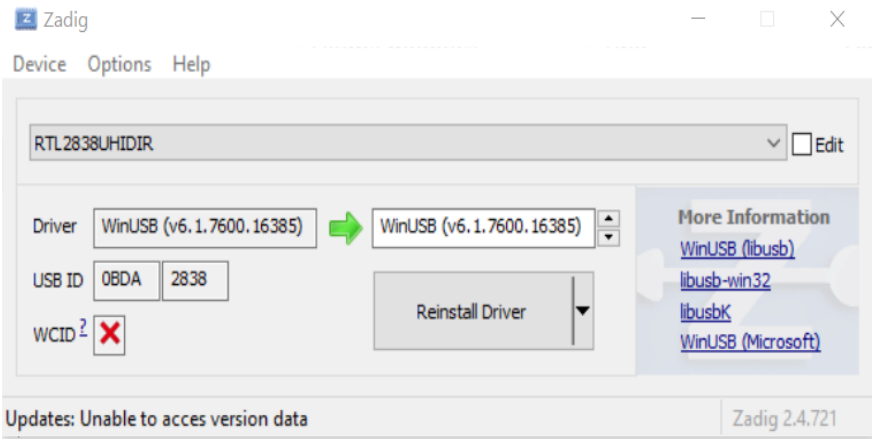
- Object Oriented Programming (OOP) allows to create classes:
  - Description of the data type structure (fields or properties)
  - The set of operations (methods or functions) defined for this data type
- In MATLAB an object is a variable belonging to a specific class: before defining an object of a class it is necessary to know well the characteristics of the class.
- The operations that can be performed on a class are restricted to the methods defined for that class.
- For almost every class defined by MATLAB there is the `step` command, whose operation changes from class to class and depends on the class itself.

Before starting…….

- Install Support Package for RTL-SDR Radio
  - On the MATLAB Home tab click **Add-Ons > Get Hardware Support Packages**.
  - In Add-On Explorer, browse or search for the **Communications Toolbox™ Support Package for RTL-SDR Radio**.
  - Select the support package and then click Install.
  - During support package installation, you will be prompted to install the drivers needed for the RTL-SDR Radio software.

How to install drivers for RTL-SDR in MATLAB

- After having installed the **Communications Toolbox Support Package for RTL-SDR Radio**
- Open **File Explorer** on your PC.
- Follow this path: C:\ProgramData\MATLAB\SupportPackages\<version>\3P.instrset\zadig.instrset\zadig\zadig-XX.exe
  - <version> is the MATLAB version
  - XX is the Zadig version number
- Connect the RTL-SDR to the PC
- Open **Zadig** application, select RTL from the list
- Install **Driver**



Hardware Setup

1. Plug the RTL-SDR into your computer
2. Start MATLAB, at the MATLAB command prompt, call the `sdrsetup` function.
3. To get information for all radios connected to your computer, call the `sdrinfo` function.

```
        hwinfo = sdrinfo
hwinfo =
        RadioName: 'Generic RTL2832U OEM’
        RadioAddress: ‘0’
        RadioIsOpen: 0
        TunerName: 'R820T’
        Manufacturer: 'Realtek’
        Product: 'RTL2838UHIDIR’
        GainValues: [29×1 double]
        RTLCrystalFrequency: 28800000
        TunerCrystalFrequency: 28800000
        SamplingMode: 'Quadrature’
        OffsetTuning: 'Disabled'
```

Load RTL-SDR driver

- Construct an RTL-SDR receiver System object:

```
obj_rtlsdr = comm.SDRRTLReceiver
```

```
obj_rtlsdr =
        comm.SDRRTLReceiver with properties:
        RadioAddress: ‘0’
        CenterFrequency: 102500000
        EnableTunerAGC: true
        SampleRate: 250000
        OutputDataType: 'int16’
        SamplesPerFrame: 1024
        FrequencyCorrection: 0
        EnableBurstMode: false
```

Carrier frequency of the received signal

Bandwidth of the received signal

Number of samples passed to MATLAB with each call to *step* function